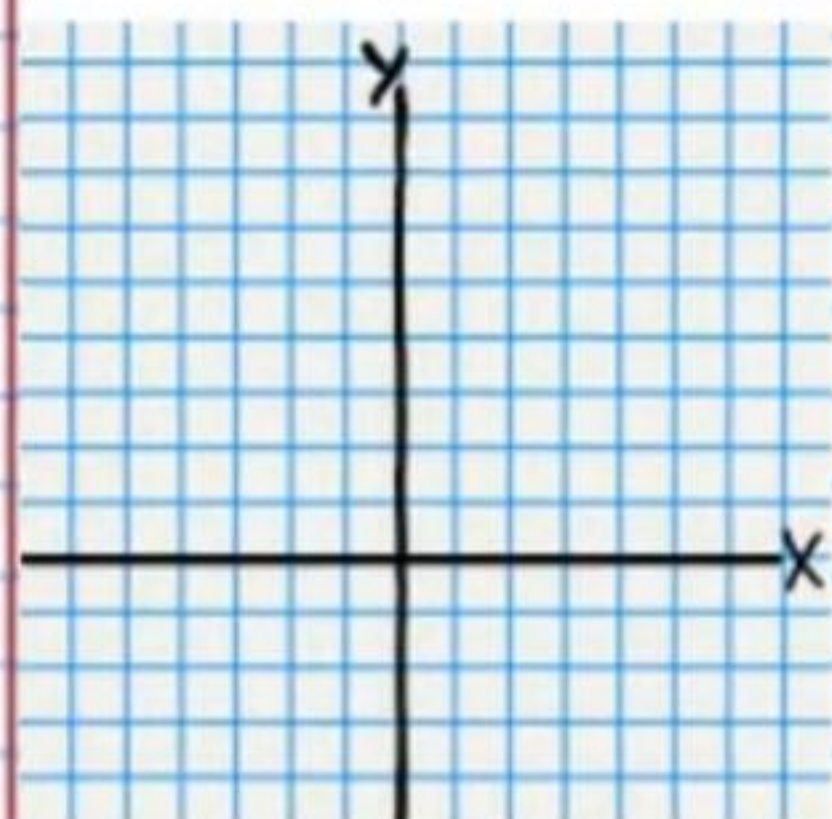


Parametric Equations and Their Graphs (Homework)

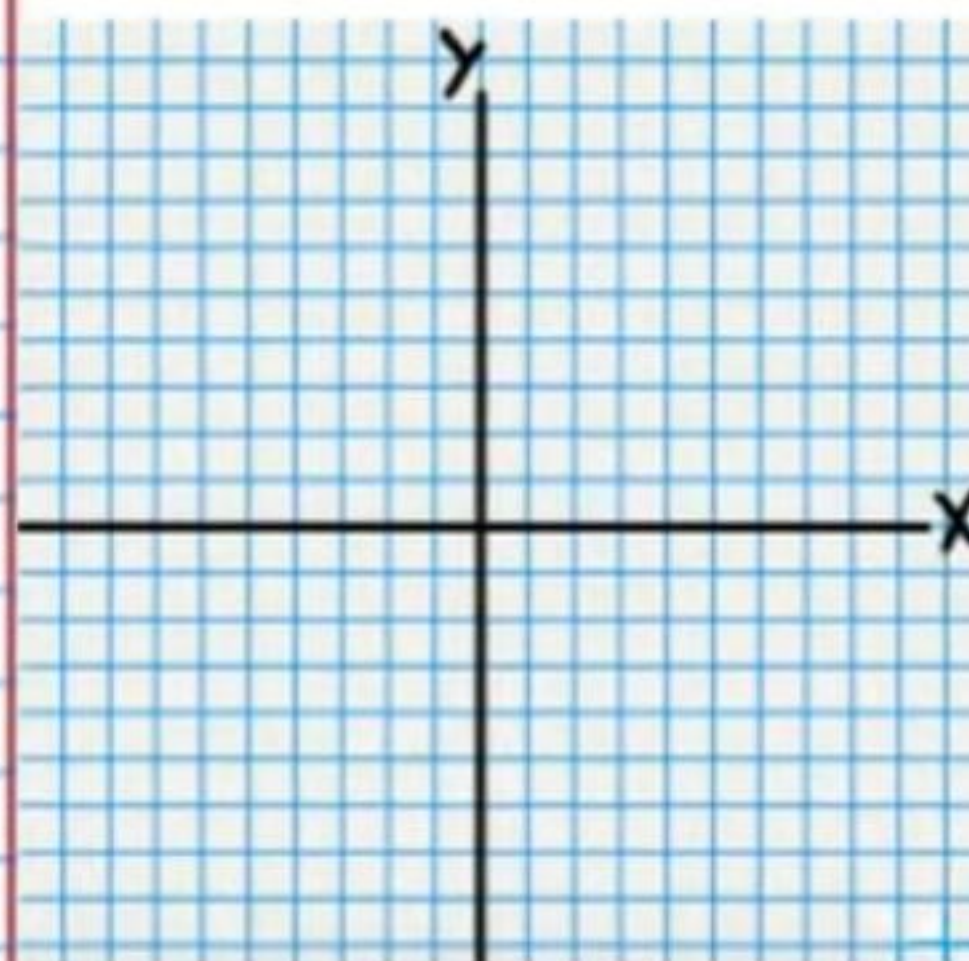
Graph the following equations. Write the coordinates of at least 4 points. Use a graphing calculator to draw the other parts of the plot. Then eliminate the parameter.

• ① $x = t$, $y = 2 + t^2$ Point #1 Point #2



Point #3 Point #4 Eliminate the parameter

② $x = t^3 + 1$, $y = t - 1$ Point #1 Point #2



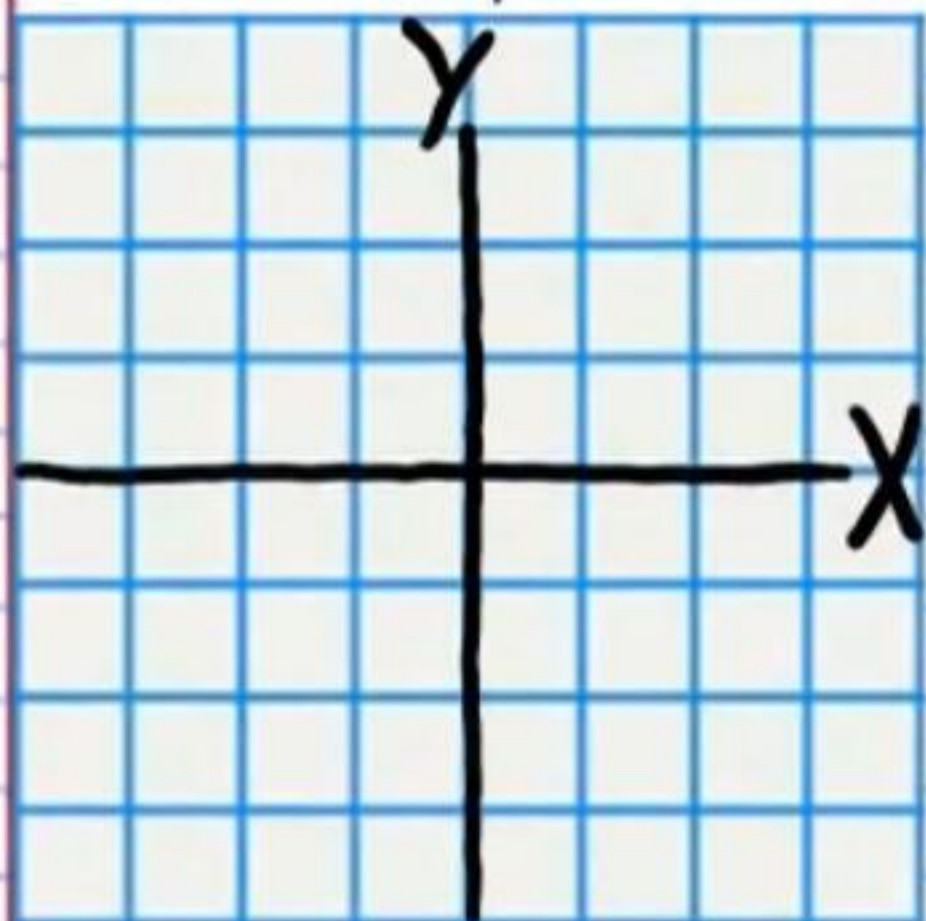
Point #3

Point #4 Eliminate the parameter

③ $x = -2t, y = t$

Point #1

Point #2



Point #3

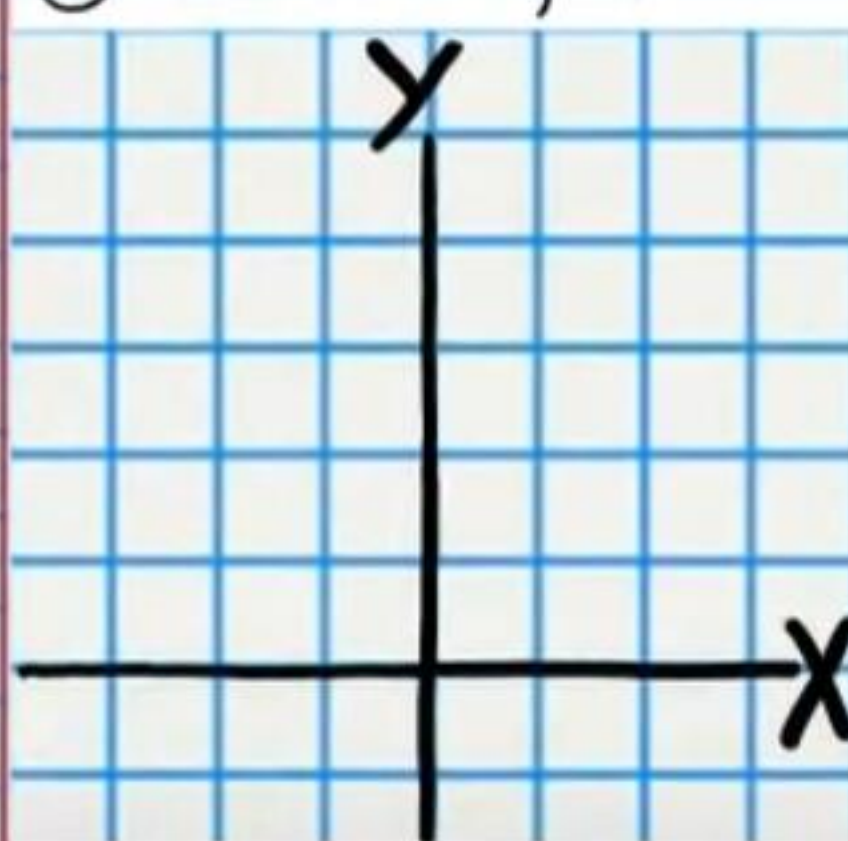
Point #4

Eliminate the parameter

④ $x = \sqrt{t+2}, y = t+3$

Point #1

Point #2



Point #3

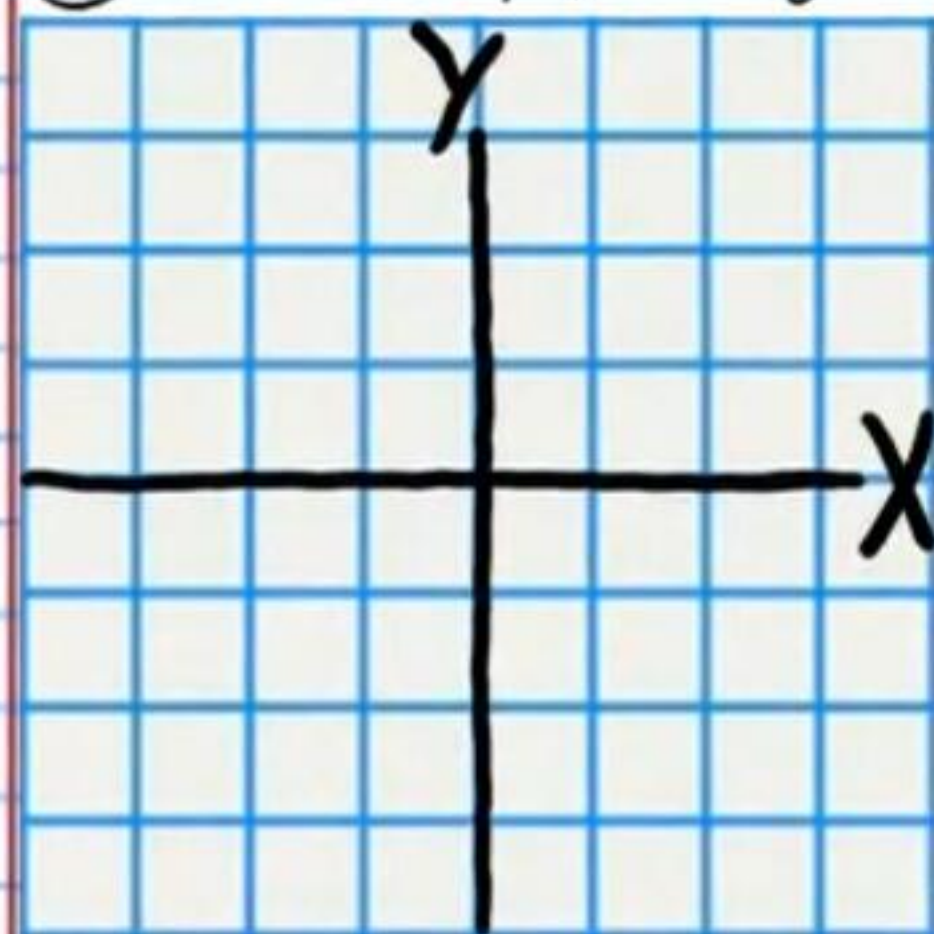
Point #4

Eliminate the parameter

⑤ $x = t - 1, y = \frac{1}{t}$

Point #1

Point #2



Point #3

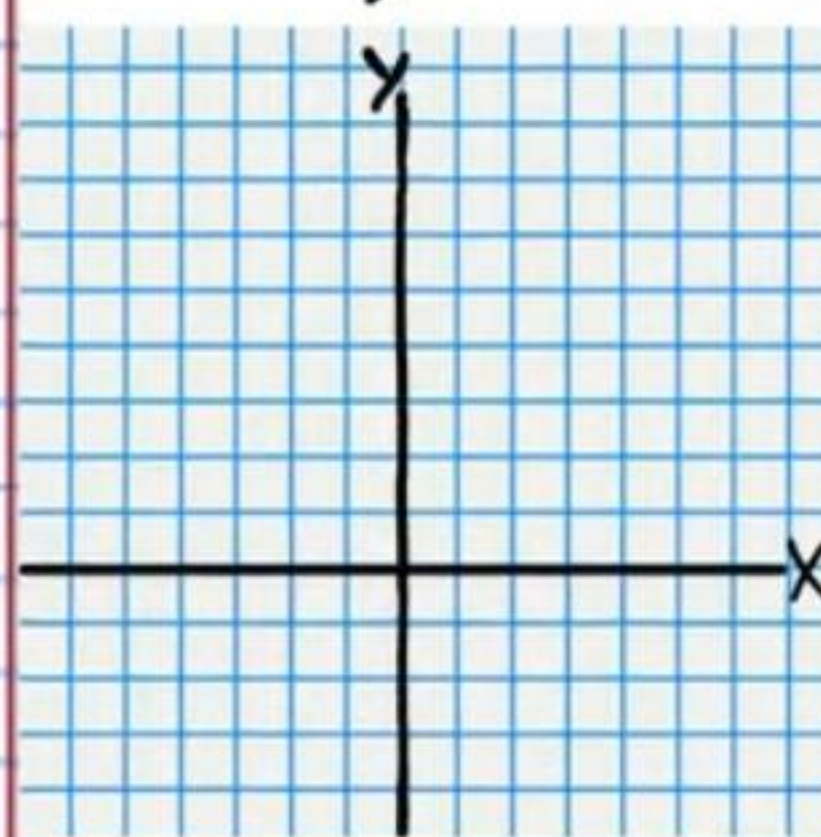
Point #4

Eliminate the parameter

⑥ $x = \frac{2^t}{3} + 1, y = 2^t$

Point #1

Point #2

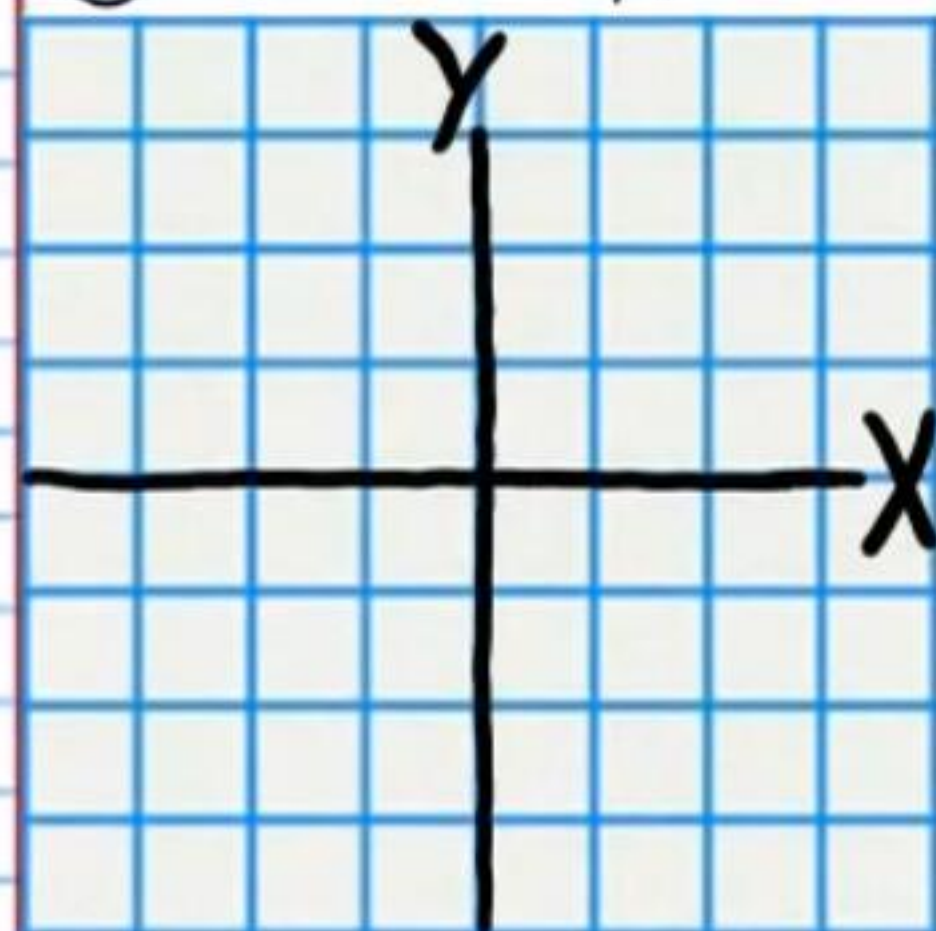


Point #3

Point #4

Eliminate the parameter

⑦ $x = \sin^2 t$, $y = \sin t$ Point #1 Point #2

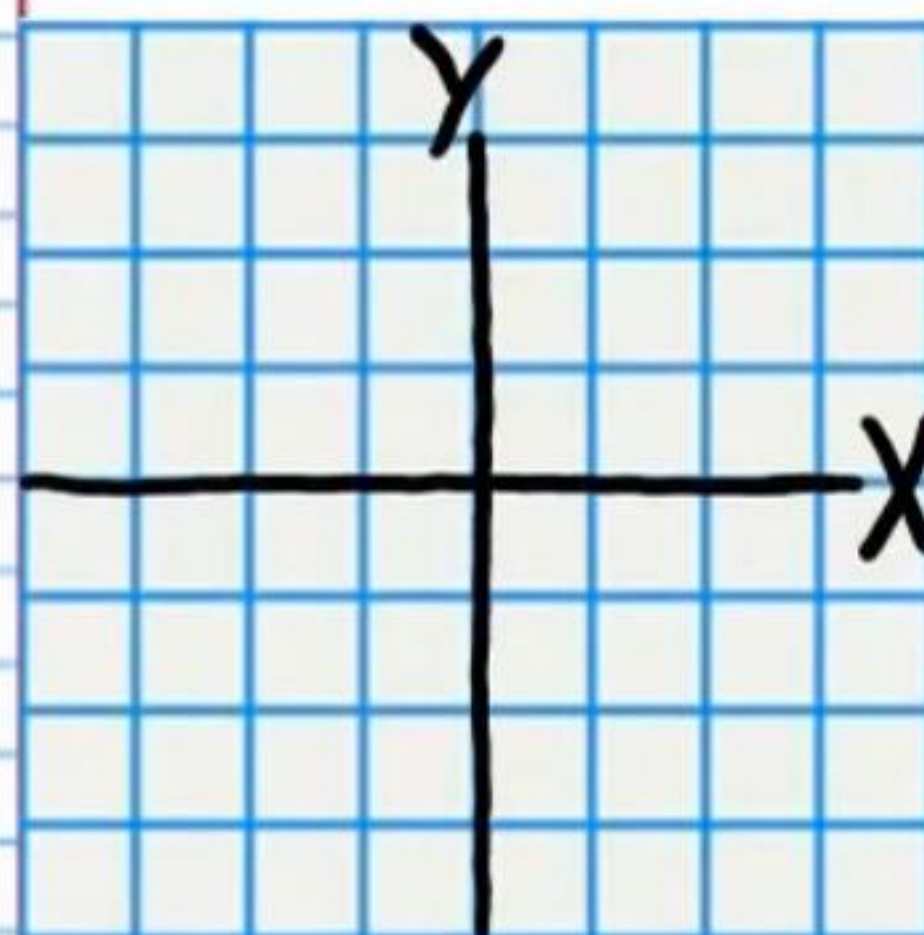


Point #3

Point #4

Eliminate the parameter

⑧ $x = 3 \cos t$, $y = 3 \sin t$ Point #1 Point #2

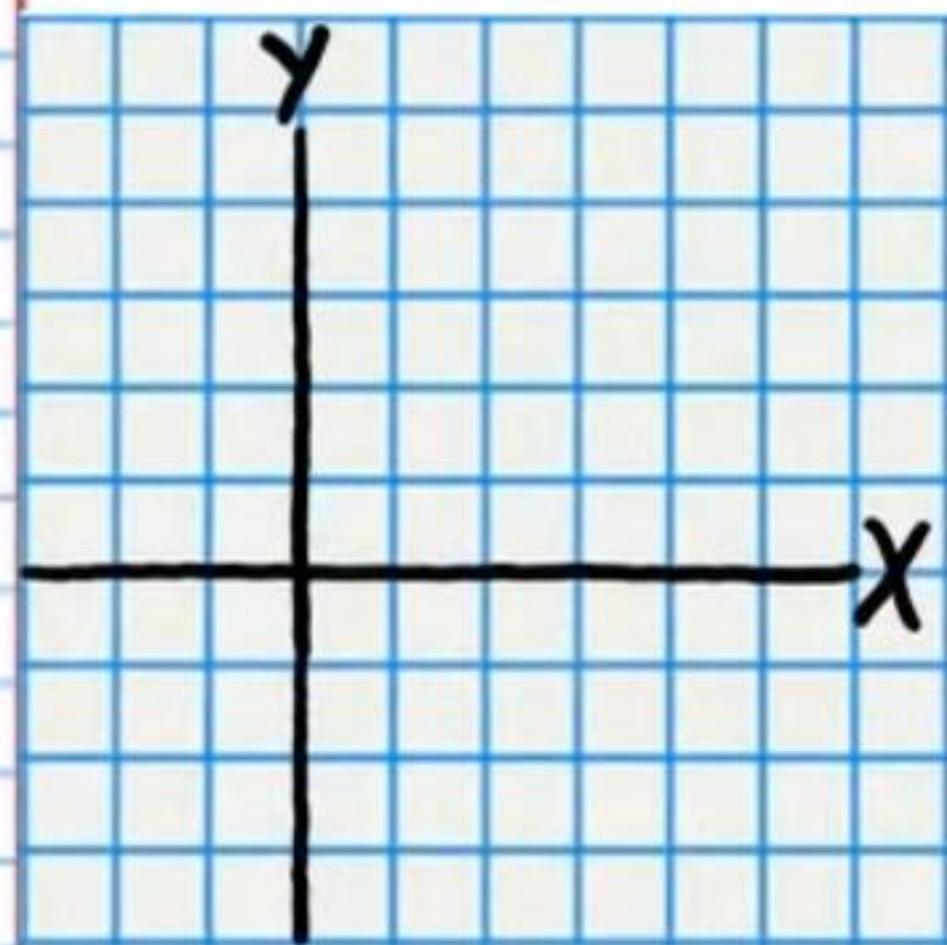


Point #3

Point #4

Eliminate the parameter

⑨ $x = 3\cos t + 2$, $y = 2\sin t + 2$ Point #1 Point #2



Point #3

Point #4

Eliminate the parameter

Create parametric equations for x and y using t as the parameter.

⑩ $y = x^3$

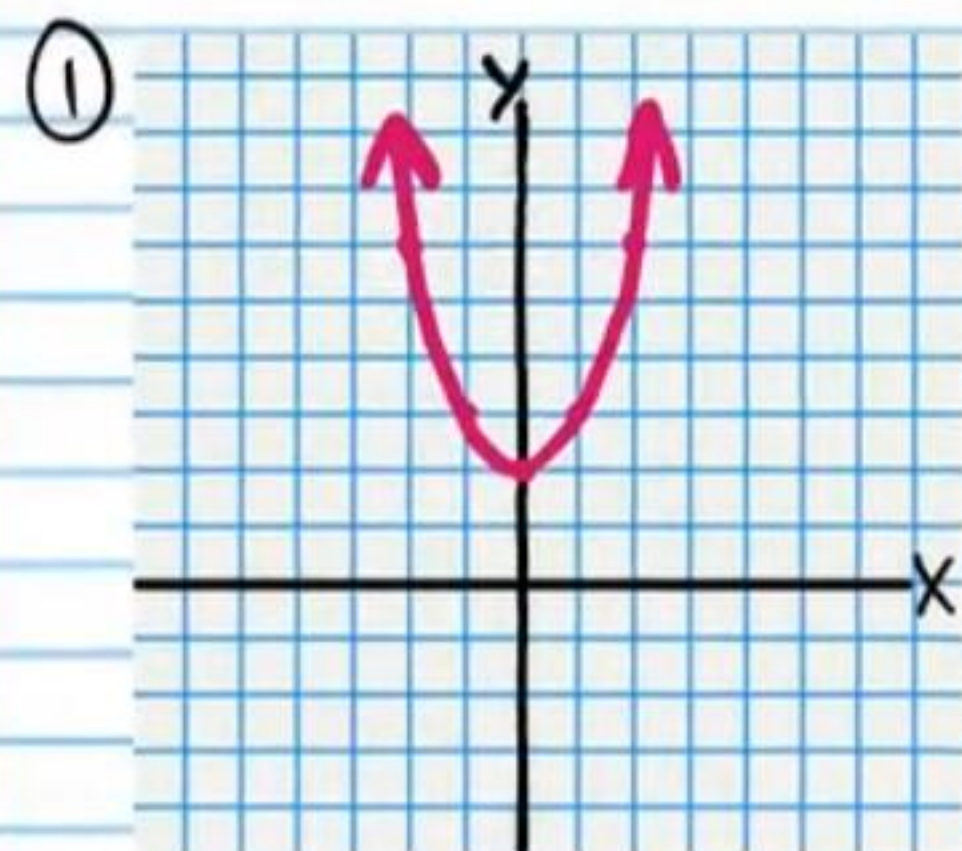
⑪ $y = \sqrt{x} - 4$

⑫ $\frac{y+3}{4} = x+2$

⑬ $\frac{x^2}{9} - y = 6$

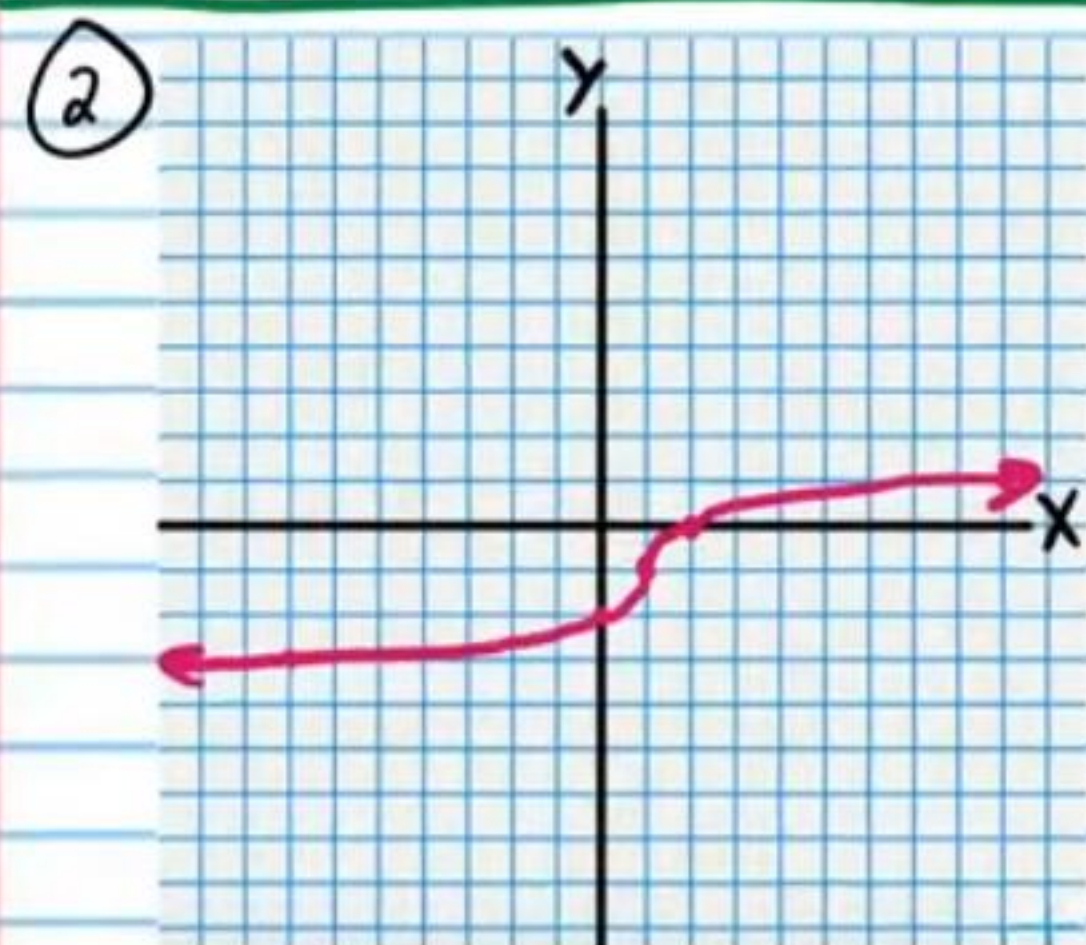
Answers

* Other points are possible.



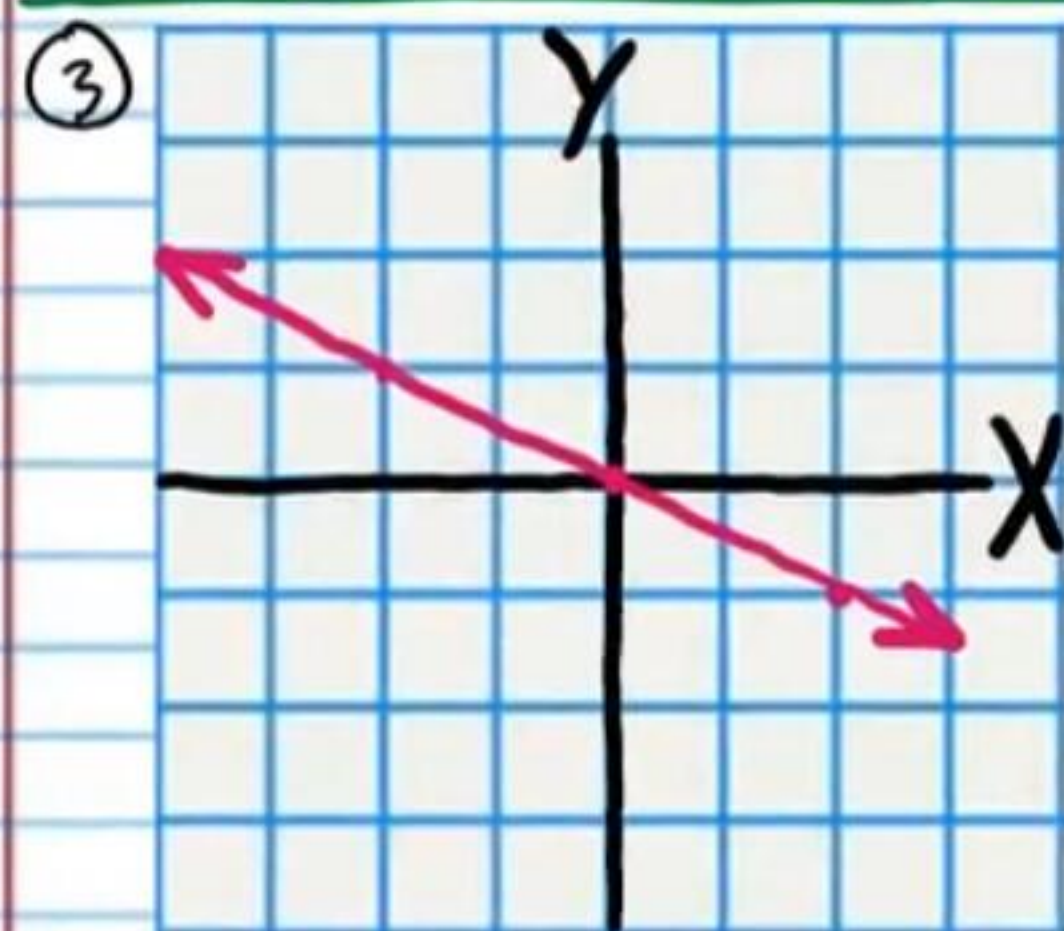
$t=0$	$t=-1$
$(0, 2)$	$(-1, 3)$
$t=1$	$t=2$
$(1, 3)$	$(2, 6)$

$$y = 2 + x^2$$



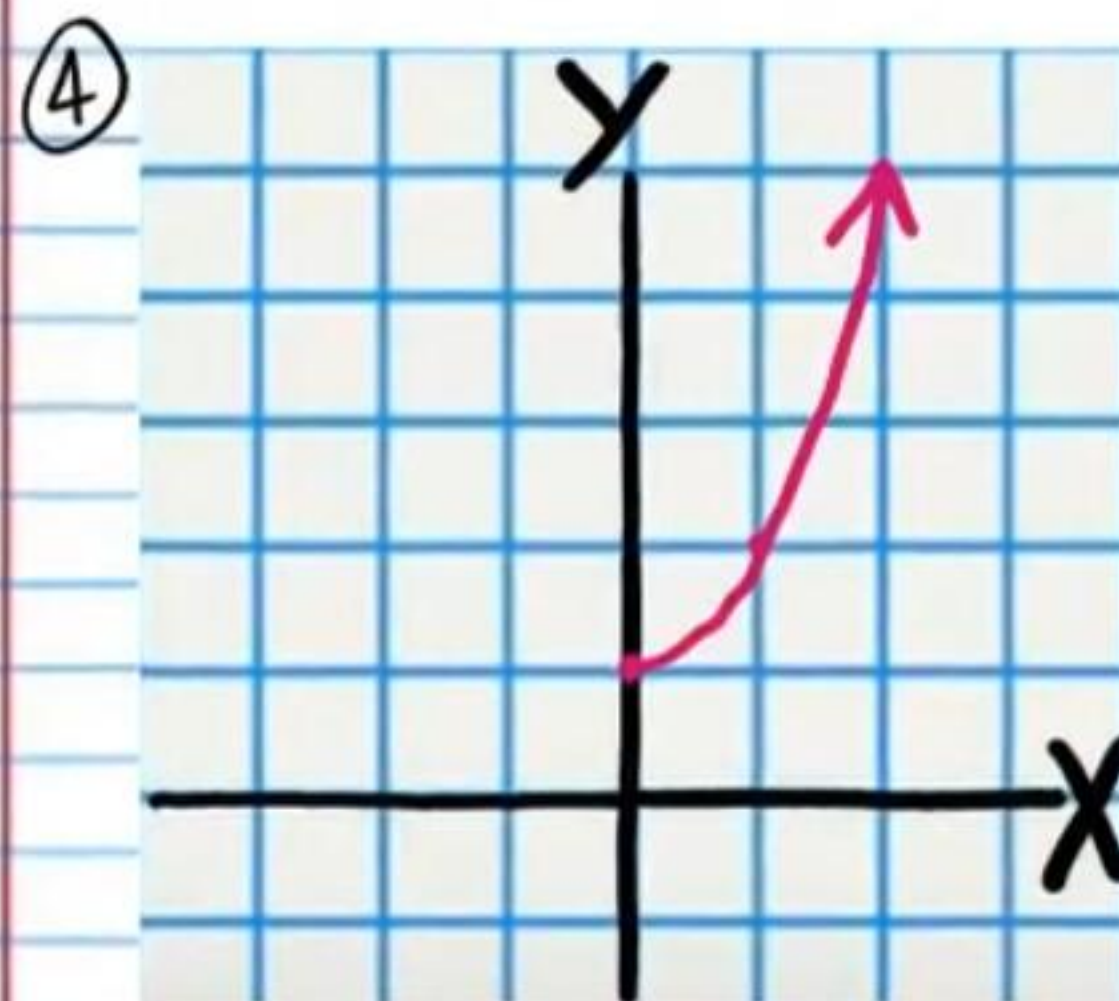
$t=0$	$t=1$
$(1, -1)$	$(2, 0)$
$t=-1$	$t=2$
$(0, -2)$	$(9, 1)$

$$x = (y+1)^3 + 1$$



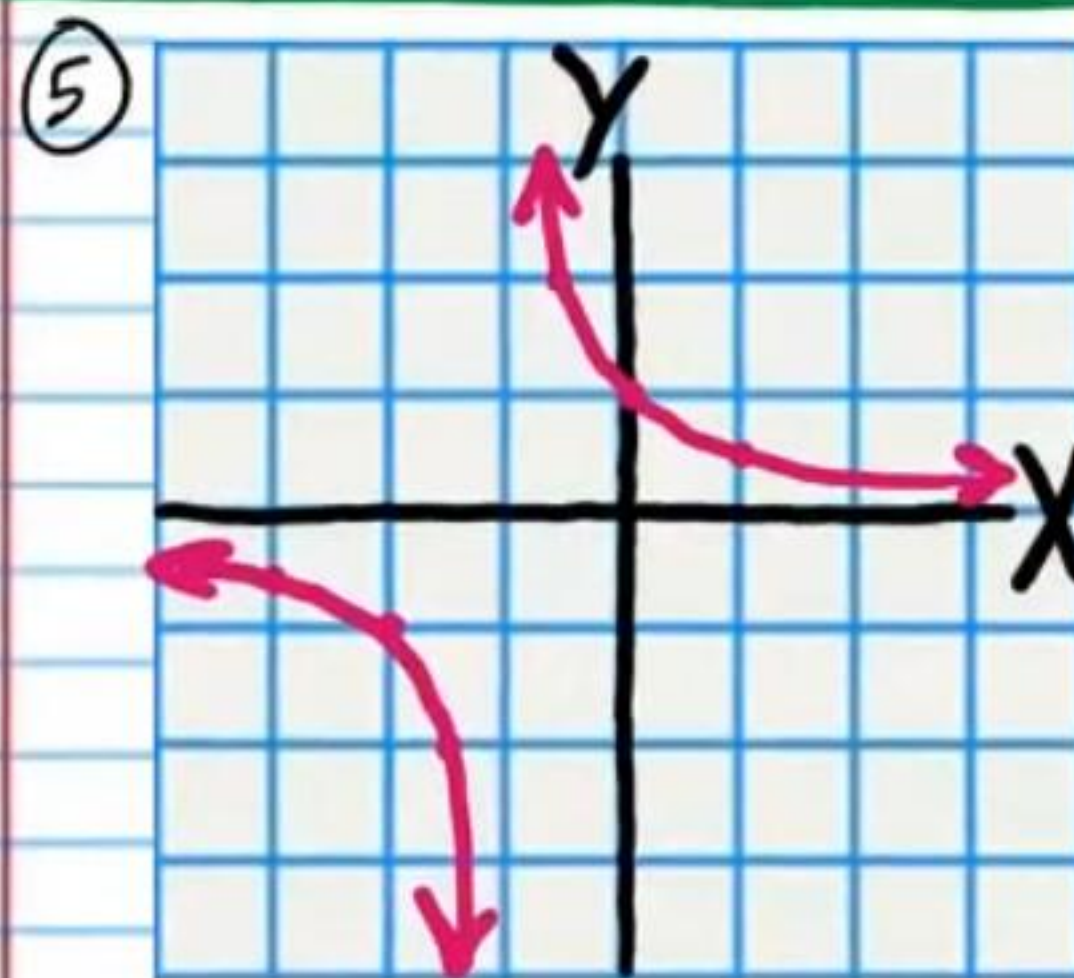
$t=0$	$t=1$
$(0, 0)$	$(-2, 1)$
$t=-1$	$t=2$
$(2, -1)$	$(-4, 2)$

$$x = -2y$$



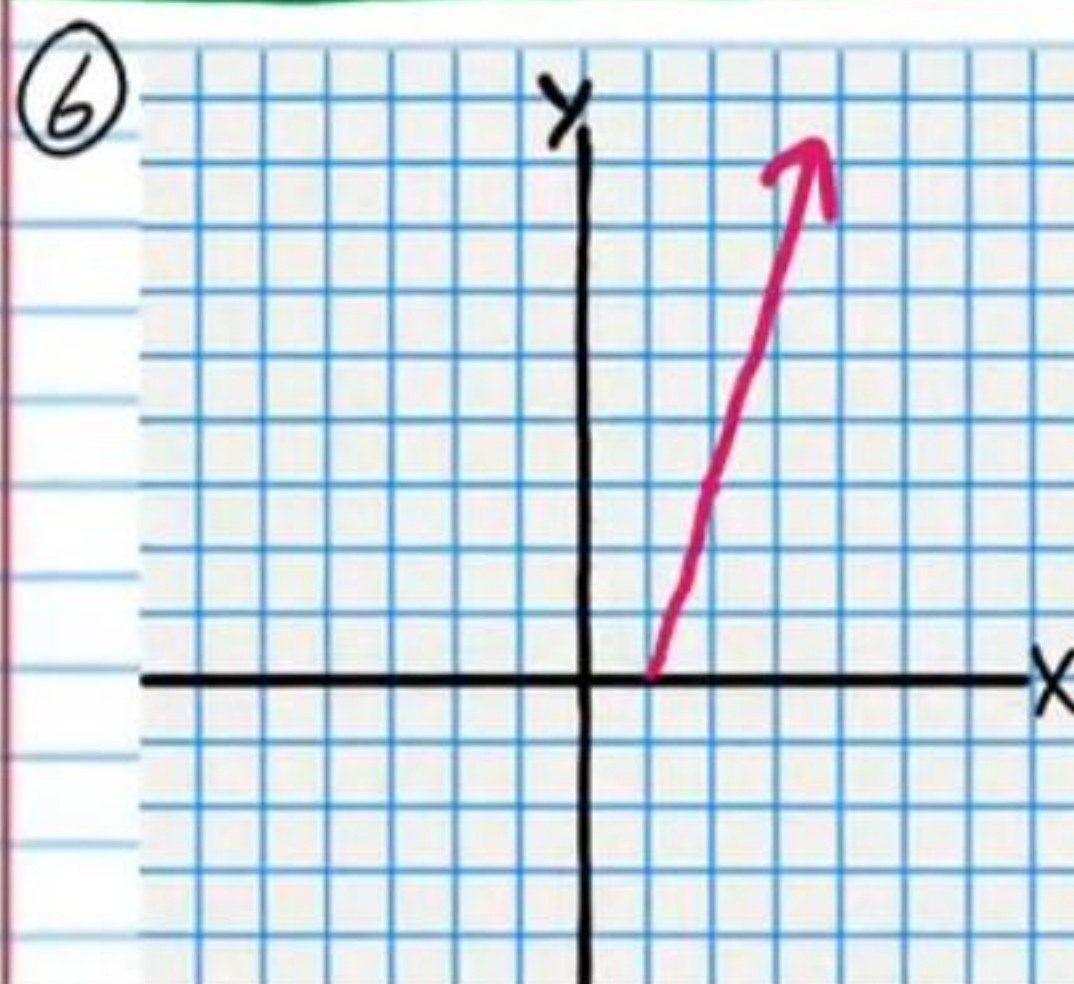
$t=-2$	$t=-1$
$(0, 1)$	$(1, 2)$
$t=2$	$t=0$
$(2, 5)$	$(\sqrt{2}, 3)$

$$x = \sqrt{y-1}$$



$t=1$	$t=2$
$(0, 1)$	$(1, \frac{1}{2})$
$t=-1$	$t=-2$
$(-2, -1)$	$(-3, -\frac{1}{2})$

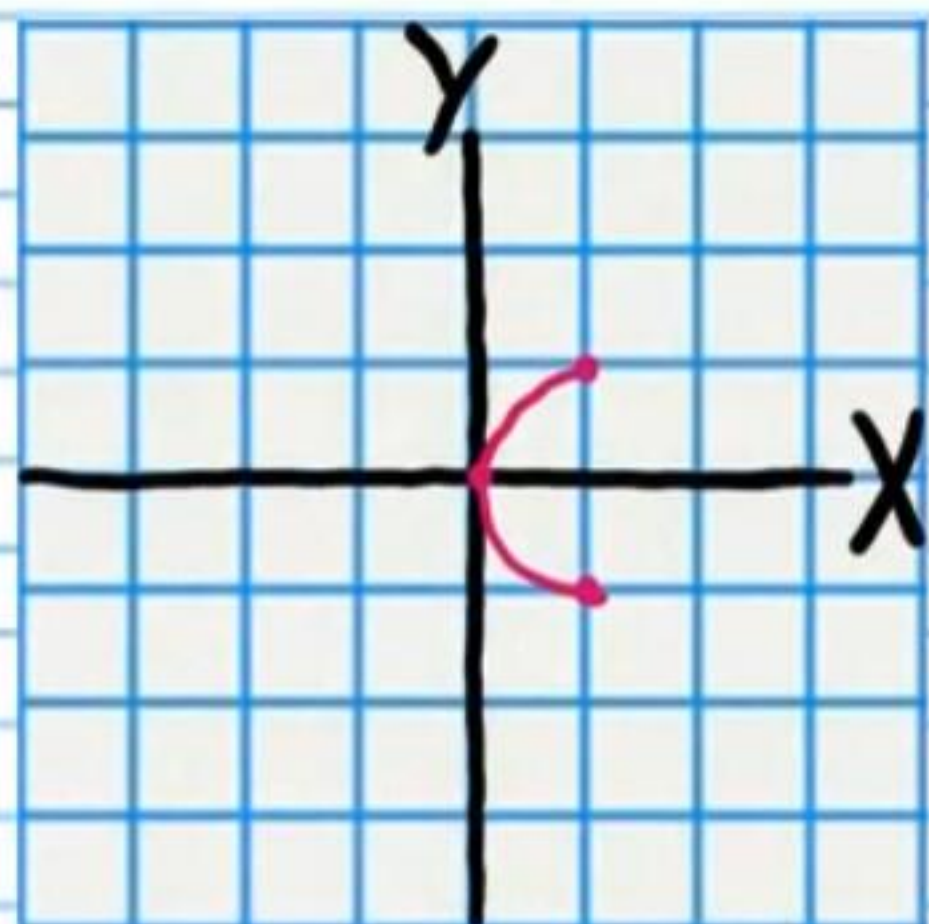
$$y = \frac{1}{x+1}$$



$t=0$	$t=1$
$(1\frac{1}{3}, 1)$	$(1\frac{2}{3}, 2)$
$t=2$	$t=3$
$(2\frac{1}{3}, 4)$	$(3\frac{2}{3}, 8)$

$$x = \frac{y}{3} + 1$$

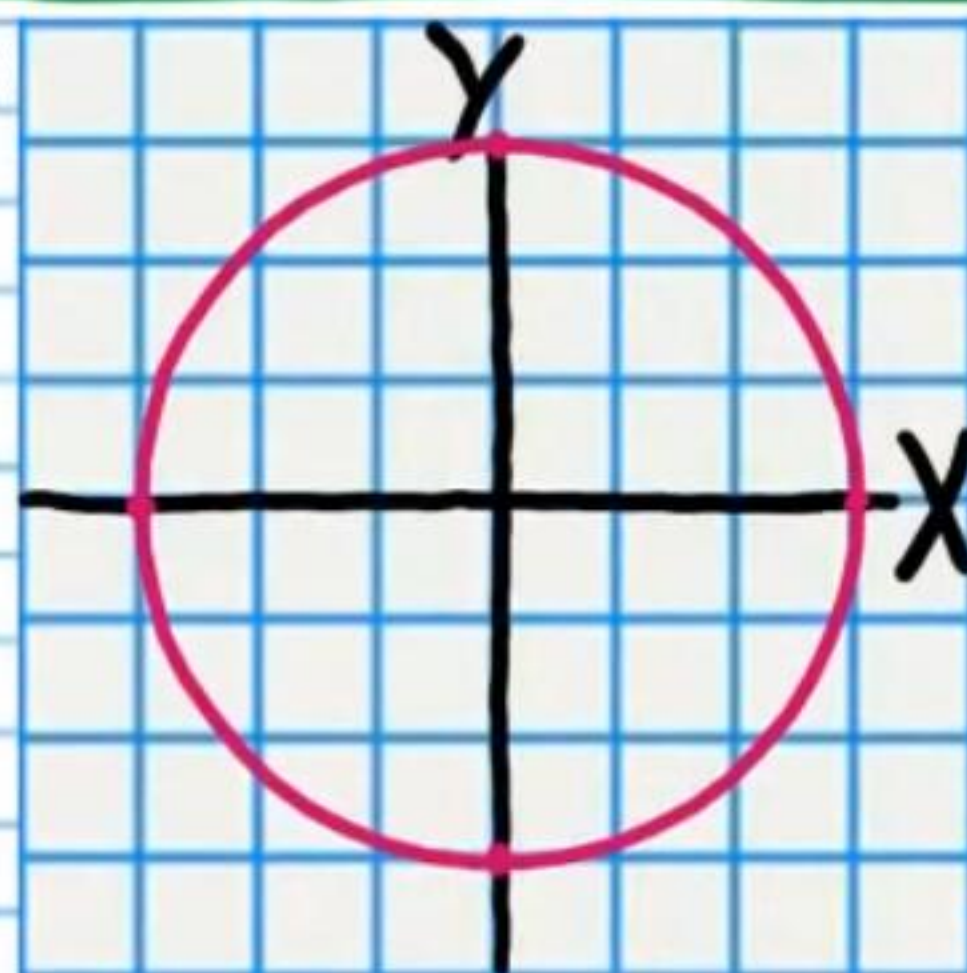
⑦



$t = 0^\circ$	$t = -90^\circ$
$(0, 0)$	$(1, -1)$
$t = 90^\circ$	$t = 45^\circ$
$(1, 1)$	$(\frac{1}{2}, \frac{\sqrt{2}}{2})$

$x = y^2$

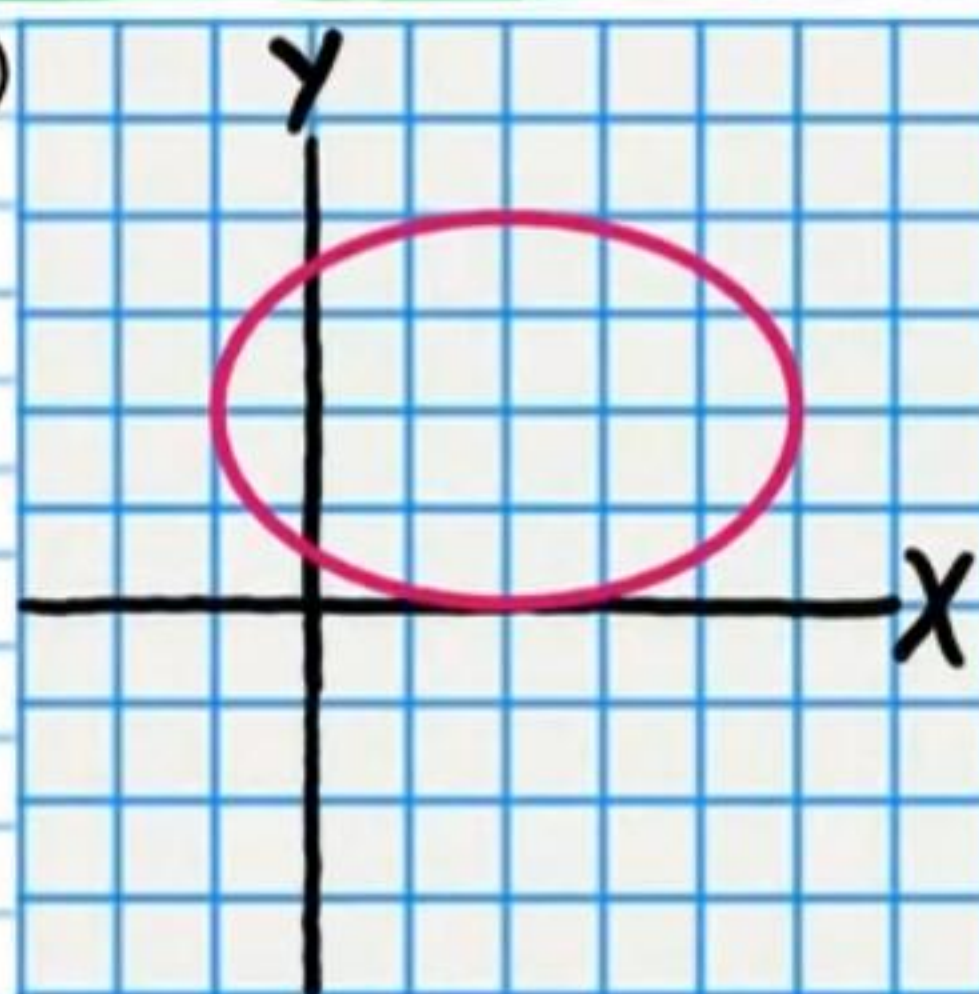
⑧



$t = 0^\circ$	$t = 90^\circ$
$(3, 0)$	$(0, 3)$
$t = -90^\circ$	$t = 180^\circ$
$(0, -3)$	$(-3, 0)$

$x^2 + y^2 = 9$

⑨



$t = 0^\circ$	$t = -90^\circ$
$(5, 2)$	$(2, 0)$
$t = 90^\circ$	$t = 180^\circ$
$(2, 4)$	$(-1, 2)$

$\frac{(x-2)^2}{9} + \frac{(y-2)^2}{4} = 1$

* Other equations are possible.

⑩

$x = t, y = t^3$

⑪

$x = t, y = \sqrt{t} - 4$

⑫

$x = t, y = 4t + 5$
or
 $y = 4(t+2) - 3$

⑬

$x = t, y = \frac{t^2}{9} - 6$